

Legume cover crops differentially alter soil organic carbon fractions across species and over time:
microbial-mediated mechanisms and carbon sequestration implications

Lin Wei ^a, Runze Zhang ^a, Wenhai Wang^c, Yueyan Wang ^a, Pengwei Yao ^b, Shanshan Zhang ^{c*}

^a College of Forestry, Henan Agricultural University, Zhengzhou 450002, China

^b College of Tobacco Science, Henan Agricultural University, Zhengzhou 450002, China

^c Institute of Vegetables, Tibet Academy of Agricultural and Animal Husbandry Sciences, Lhasa,
Tibet 850002, China

Abstract

Legume cover crop represents a promising agricultural practice for enhancing soil organic carbon (SOC) sequestration. However, their species-specific SOC sequestration potential and underlying mechanisms in orchard ecosystems remain incompletely understood. We examined high-altitude orchards in Tibet, China, using SOC fractionation, metagenomic sequencing, and biomarker analysis. Our objectives were to assess legume cover crop impacts on SOC fractions and microbial properties, and uncover species-specific SOC sequestration mechanisms. Short-term cover by *Trifolium repens* and *Vicia villosa* significantly reduced particulate organic carbon (POC), but minimally affected mineral-associated organic carbon (MAOC), resulting in unchanged or reduced total SOC. Long-term legume cover crops enhanced SOC sequestration through concurrent increases in both POC and MAOC, exhibiting species specificity: *T. repens* promoted greater SOC sequestration primarily through more efficient POC accumulation, whereas *V. villosa* enhanced SOC stability by facilitating the formation of MAOC. These differences were attributed to their distinct effects on microbial SOC transformation pathways. *T. repens* with lower C/N ratio and

* Corresponding author.

E-mail address: zhangshanshan@taas.org (S.S. Zhang)

higher non-structural carbohydrates stimulated soil microbial metabolism and hydrolase enzyme activity, accelerating plant residues conversion to POC via the soil microbial ex vivo pathway. Under *V. villosa* cover, elevated oxidative enzyme activities facilitated direct plant residue incorporation into MAOC, as evidenced by relative lower soil microbial necromass carbon (MNC) compared to *T. repens* treatment. Our results demonstrate species-dependent heterogeneity in how legume cover crops mediate SOC sequestration and stabilization, providing a scientific basis for optimizing cover crop strategies in marginal orchards.

Key words: Cover crop; Mineral-associated organic carbon; Amino sugars; Microbial community

1. Introduction

Soil organic carbon (SOC), the largest terrestrial carbon pool storing over 80% of Earth's land-based carbon, is critical for climate change mitigation (Schmidt et al., 2011). However, intensive agricultural practices-including tillage, monocropping, and over-fertilization have disrupted carbon cycles, converting agroecosystems from carbon sinks to net sources (IPCC, 2021). Orchards, particularly susceptible due to prolonged cultivation, limited organic inputs, and erosion, face severe SOC depletion (Martínez-Mena et al., 2020). Cultivating cover crops in orchard alleys offers a promising strategy to enhance SOC sequestration.

A global meta-analysis demonstrates that cover cropping increases soil organic carbon (SOC) by an average of 15.5% (Jian et al., 2020). Cover crops enhance soil carbon sequestration through three integrated pathways: (1) increasing organic inputs via plant residues and rhizodeposits; (2) improving microbial CUE by mitigating nutrient constraints, thereby promoting microbial-derived carbon accumulation; and (3) stimulating macroaggregate formation through root activity, which physically protects organic carbon from mineralization (Ball et al., 2022; Hu et al., 2023b).

Contemporary SOC formation framework differentiates SOC into two fractions: particulate organic carbon (POC) and mineral-associated organic carbon (MAOC) (Lavalley et al., 2020). POC, primarily originating from the physical breakdown of plant residues, has a faster turnover rates compared to MAOC, which is mainly derived from microbial necromass and stabilized through organo-mineral interactions (Cotrufo and Lavalley, 2022). It is worth noting that the total organic carbon content in soil may not change significantly in the early stages of cover cropping (Fulton-Smith and Cotrufo, 2019). However, the sources and fractions of soil organic carbon, as sensitive indicators, can quickly respond to changes in management practices and reflect the long-term stable potential of the soil organic carbon pool (Heckman et al., 2021). Analyzing SOC sources and fractions enhances the assessment of cover crops' carbon sequestration potential and clarifies the underlying mechanisms.

The microbial carbon pump (MCP) framework elucidates how microbial metabolic processes regulate SOC fraction dynamics through two interdependent mechanisms: (1) *in vivo* modification via selective substrate assimilation and microbial necromass production, and (2) *ex vivo* modification through extracellular enzymatic catalysis and metabolic byproduct excretion (Cotrufo et al., 2013; Liang et al., 2017). This metabolic dichotomy drives two counteracting stabilization outcomes: microbial necromass derived from intracellular turnover predominantly contributes to MAOC formation through organo-mineral complexation, while extracellular polymeric substances and partially decomposed plant residues preferentially stabilize as POC within macroaggregates, ultimately governing the net soil carbon balance (Liang et al., 2017). Cover crop types differing in chemical composition and rhizodeposition patterns drive variations in organic carbon input quantity/quality, thereby eliciting divergent microbial metabolic activity that consequently regulates

SOC dynamics (Canisares et al., 2023; Hu et al., 2023a; Zhou et al., 2023). The soil MCP offers mechanistic insights into how these variations govern SOC accumulation and stabilization across cover crop systems, while highlighting its critical role in understanding carbon cycling under sustainable agricultural practices.

Orchards in Tibet are vulnerable to SOC loss due to inherently low native SOC, excessive chemical fertilizer use, and harsh climates. *Vicia villosa* R. and *Trifolium repens* L. are dominant legume cover crops there, selected for cold tolerance, rapid growth, and high nitrogen fixation. While their short-term soil fertility benefits are known, the effects on SOC sequestration and stabilization in local orchards remain unclear. Therefore, we conducted a study in Linzhi, Xizang (Tibet), integrating a dataset of SOC physical fractions, enzyme activities, microbial communities, and microbial necromass. We aimed to determine: (1) which cover crop, *Vicia villosa* R. or *Trifolium repens* L., enhances SOC sequestration and stability more effectively in orchard soil, and (2) the underlying mechanisms. We hypothesized that: a) *Trifolium repens* L. with higher nitrogen and non-structural carbohydrate, would stimulate microbial pathways promoting MAOC sequestration; and b) *Vicia villosa* R. with higher cellulose and lignin content, would reduce the efficiency of the microbial in vivo pathway for SOC turnover, favoring POC accumulation. This study expands our understanding of how cover crops regulate microbial transformation pathways to optimize SOC sequestration.

2. Materials and Methods

2.1. Experimental sites

The study was conducted in the cherry orchard at Gama Farm (29°22'02"N, 94°31'14"E; 2,880 m asl) in Bayi District, Nyingchi City, Tibet. This region experiences the temperate humid monsoon

climate, with the mean annual precipitation is 654 mm and the average temperature of 8.5°C (-2°C in January to 20°C in July). More than 90% of the annual rainfall is concentrated from May and September. The soil is categorized as yellow earth (Haplic Acrisols, FAO Soil Taxonomy System).

The orchard has adopted *Vicia villosa* (cultivation from 2018 and 2023) and *Trifolium repens* (cultivation from 2019 and 2024) as cover crops, deployed to enhance soil organic matter accumulation and improve soil fertility. The cover crops were spring-sown in the orchard alleyway, not fertilized and allowed to senesce naturally, with aboveground biomass retained in situ. Fertilizer was applied to the cherry trees following local farming practices, the specifics of which are detailed in Table S1.

2.2 Experimental Design

The experiment was conducted in a randomized, fully complete block design. Five treatments were established in five replicates: short-term *V. villosa* cultivation from 2024 (SV), long-term *V. villosa* from 2020 (LV), short-term *T. repens* from 2024 (ST) and long-term *T. repens* from 2020 (LT), with bare soil without cover crops as control (CK). In September 2024, the aboveground parts of cover crops were cut at ground level in a 1 m² quadrats per plot and oven dried at 70 °C for 48 h for to calculate the aboveground biomass. Then the senesced plant residues were ground in a cyclone mill with a filter mesh of 1 mm sieve for subsequent chemical composition analysis.

In each plots, five randomized points between the tree alleyways were sampled at depth of 0–20 cm were collected to form a composite soil sample, resulting in a total of 25 composite samples. Soil samples were cooled with ice packs prior to transport to the laboratory. In the laboratory, all visible roots, stones, and other debris in the soil samples were removed. The sieved soil samples (2-mm sieve) were divided into three subsamples, one of which was stored at 4 °C and analysed within

one week after collection to determine the soil dissolved organic C (DOC) and microbial biomass C and N (MBC and MBN) and enzyme activities. The second subsample was air-dried to determine the soil physical properties and amino sugars. The third subsample was stored at -20°C for metagenomic analysis.

2.3 Measurement protocols

2.3.1 Plant chemical properties

The total C, N and P concentration of plant residues were respectively determined by dry combustion (Nelson and Sommers, 1982), Kjeldahl (Bremner and Mulvaney, 1982) and acid-soluble-molybdenum-antimony-resistant colorimetric methods (Nelson and Sommers, 1982). Soluble sugars, starch and NSC (sum of soluble sugars and starch) concentrations were determined by the modified phenol sulphuric acid method (Zhang et al., 2014).

2.3.2 Soil chemical properties

Soil organic matter (SOM) was measured by the $\text{K}_2\text{CrO}_7\text{-H}_2\text{SO}_4$ oxidation method (Bremner, 1996). Soil available nitrogen ($\text{NH}_4^+\text{-N}$ and $\text{NO}_3^-\text{-N}$) was determined using the potassium chloride extraction-flow injection analysis method (Alef and Nannipieri, 1995). Available phosphorus (AP) was determined using the sodium bicarbonate extraction-molybdenum-antimony-resistant colorimetric method. The dissolved organic carbon (DOC) was extracted with deionized water then determined by a TOC-TN analyser. Soil microbial biomass carbon (MBC) and nitrogen (MBN) were determined using the chloroform fumigation method (Vance et al., 1987).

2.3.3 Physical fractions of soil organic carbon

Soil organic matter was fractionated into particulate organic carbon (POC, 53-2000 μm) and mineral-associated organic carbon (MAOC, $<53\text{ }\mu\text{m}$) through density-based particle size separation

(Cambardella and Elliott, 1992). Briefly, airdried bulk soil (30 g; < 2 mm; unground) was shaken in 90 mL of sodium hexametaphosphate solution (5 g L⁻¹) at 120 rpm for 18 h. The homogenized suspension was sequentially fractionated using a 53-μm stainless-steel sieve, with the retained POC fraction rinsed repeatedly with deionized water until the rinse water became colourless. The fraction that passed through the sieve (<53 μm) was collected after 48 h gravitational settling. All separated fractions were oven-dried at 40 °C until a constant weight was reached. POC and MAOC were analysed using the K₂CrO₇–H₂SO₄ oxidation method.

2.3.4 Soil amino sugars

Amino sugars were analyzed following the method established by Zhang and Amelung (1996). Briefly, a 1 g freeze-dried soil sample (<0.153 mm) was hydrolysed for 8 h at 105 °C in 6 M HCl (10 mL). The resulting hydrolysate was filtered and evaporated to dryness at 50–53 °C under reduced pressure and redissolved in 5 mL of MilliQ water. The residue was adjusted to 6.6–6.8 using 1 M KOH and 0.01 M HCl. The hydrolysate samples were centrifuged, and the supernatant was freeze-dried; the residue was then washed with methanol, and the recovered amino sugars were transformed into aldononitrile derivatives detected using an gas chromatography (Agilent, Santa Clara, USA). The content of total amino sugars was calculated via summing up glucosamine (GluN), galactosamine (GalN), amino mannanose (ManN) and muramic acid (MurN).

Bacterial necromass-C (μg g⁻¹) was calculated as follows:

$$\text{Bacterial necromass-C} = \text{MurN} \times 45$$

Fungal necromass-C was determined by subtracting bacterial GluN from total GluN, assuming that MurA and GluN occur at a 1–2 M ratio in bacterial cells:

$$\text{Fungal necromass-C} = (\text{mmol GluN} - 2 \times \text{mmol MurN} \times 179:2 \times 9$$

where 179.2 is the molecular weight of GluN, and 9 is the conversion value of fungal GluN to fungal C. The total microbial necromass was estimated as the sum of fungal-derived C and bacterial-derived C. The ratio of MNC to microbial biomass carbon defines the necromass accumulation coefficient (Wang et al., 2021).

2.3.5 Soil extracellular enzyme activity

Extracellular enzyme of four hydrolytic (β -glucosidase (BG), β -N-acetylglucosaminidase (NAG), cellulose hydrolase (CBH) and alkaline phosphatase (ALP)) and two oxidase activities (polyphenol oxidase (PPO) and peroxidase (POD)) were measured according to a fluorometric method modified from Saiya-Cork et al. (2002). Enzymes activities were using 96-well microplates and reading the fluorescence on a microplate reader (BioTek Synergy HT MultiMode, BioTek Instrument, USA). These enzymes have distinct functions: BG for carbohydrate degradation, CBH for cellulose degradation, NAG for chitin degradation and AP for simple phosphomonoester degradation, PPO and POD for phenolic and lignin compounds degradation.

2.3.6 Soil microbial community structure and composition

The composition and structure of soil microbial communities were characterized via metagenomic sequencing. Samples collected from various soil either as original specimens or extracted DNA are shipped to our company under frozen conditions ($\leq 0^{\circ}\text{C}$). Upon receipt, all samples undergo rigorous quality control. Qualified DNA samples proceed to library preparation, followed by library quality assessment. Libraries passing QC are sequenced using the Illumina platform (PE150 strategy). The resulting raw sequencing data (Raw Data) are then used for downstream bioinformatic analysis. Microbial diversity metrics (Shannon index, Pielou's evenness) and compositional profiles (relative abundance at phylum/genus level) were derived from the final

report.

2.4. Statistical analysis

One-way analysis (ANOVA) was performed to test the effects of cover crops on SOC fractions and microbial properties. If significant differences were found ($p < 0.05$), means comparisons were conducted using the least significant difference (LSD) test. We used Spearman correlation analysis to examine the relationships between microbial necromass carbon and enzyme activities, as well as the relationships between soil organic carbon fractions and microbial properties (including soil microbial carbon and nitrogen, enzyme activities, microbial community diversity indices and microbial necromass carbon). Redundancy analysis (RDA) was then employed to reveal the covariance of SOC fractions and microbial properties. Finally, we used partial least squares path modelling (PLS-PM) to investigate the multivariate relationships linking SOC, soil properties, and microbial properties. All statistical analyses were performed using R version 4.1.2 (R, Inc., Seattle, USA).

3. Results

3.1 Cover crop biomass and chemical properties

Long-term planted *V. villosa* exhibited the highest biomass among treatments, while *T. repens* showed the lowest (Table 1). Short-term planted *V. villosa* had significantly higher TC and Starch concentrations than other treatments. Long-term planted *V. villosa* showed significantly higher TP, cellulose, and lignin concentrations, but significantly lower C/P ratio compared to other treatments. Long-term planted *T. repens* exhibited significantly higher TN and soluble sugar concentrations, alongside a significantly lower C/N ratio than other treatments (Table 1).

3.2 Soil properties

All cover crop treatments except short-term *V. villosa* significantly increased soil DOC, MBC

and MBN concentrations, with long-term *T. repens* showing the most pronounced effects (Table 2).

Cover by *T. repens* (both long- and short-term) significantly elevated soil NO_3^- -N, NH_4^+ -N and AN

levels. In contrast, only long-term *V. villosa* cover significantly increased soil NH_4^+ concentration.

Compared to the control, both long-term cover by *T. repens* and *V. villosa* significantly increased

soil AP contents, with *T. repens* exhibiting the stronger enhancement (Table 2).

3.3 Soil organic carbon fractions and soil properties

Under short-term treatments, *V. villosa* and *T. repens* significantly decreased soil POC by 38.24%

and 44.50%, respectively ($P < 0.05$), while MAOC showed no significant change ($P > 0.05$ for slight

decreases under *V. villosa* and increases under *T. repens* treatments) (Fig.1a). Consequently, short-

term *V. villosa* significantly reduced SOC by 17.12% ($P < 0.05$) whereas *T. repens* had no significant

effect, though the MAOC/POC ratio significantly increased by 58.31% under *T. repens* ($P < 0.05$)

with no significant change under *V. villosa* (Fig.1b).

For long-term treatments versus control, *T. repens* significantly increased both POC (+90.00%)

and MAOC (+54.96%) ($P < 0.05$), while *V. villosa* significantly increased MAOC (+72.55%, $P <$

0.05) but not POC, leading to SOC increases of 52.33% and 90.22% for *V. villosa* and *T. repens*,

respectively ($P < 0.05$) (Fig.1a). Finally, long-term *V. villosa* significantly increased the

MAOC/POC ratio (+89.84%, $P < 0.05$), contrasting with *T. repens* which significantly decreased it

(-54.19%, $P < 0.05$) (Fig. 1b).

3.4 Soil amino sugar and microbial necromass

Under short-term treatments, *V. villosa* significantly decreased soil total amino sugar (-12.83%,

$P < 0.05$) (Fig. 2a) and fungal necromass carbon (-37.05%, $P < 0.05$), while showing no significant

effect on bacterial necromass carbon (Fig. 2b). Consequently, *V. villosa* significantly reduced MNC

by 27.49% ($P < 0.05$) and the fungal-to-bacterial necromass carbon ratio by 25.68% (Fig. 2b and c).

In contrast, *T. repens* exhibited no significant effects on soil amino sugars, MNC, or their bacterial and fungal components (Fig. 2a,b and c).

Under long-term treatments, both *V. villosa* and *T. repens* significantly increased total amino sugars (+58.98% and +91.68%) and fungal necromass carbon (+81.78% and +105.7%, respectively; $P < 0.05$), contrasting with non-significant changes in bacterial necromass carbon (Fig. 2a,b). Consequently, both species significantly increased soil MNC (+54.98% and +74.97%) (Fig. 2b), and the fungal-to-bacterial necromass carbon ratio (92.65% and 68.96%, respectively; $P < 0.05$) (Fig. 2c), while the MNC-to-SOC ratio remained unaffected (Fig. 2d).

3.5 Soil enzyme activity and microbial community

Compared to the control, long-term cover crops significantly increased BG and CBH activities, whereas short-term treatments showed no significant effects (Fig. 3a and e). Long-term *V. villosa* specifically enhanced NAG and ALP activities (Fig. 3b and c), with no significant changes observed in other treatments. While peroxidase activity remained unaffected by all cover crops (Fig. 3g), polyphenol oxidase activity was significantly altered: long-term *T. repens* decreased polyphenol oxidase activity, while other treatments increased it (Fig. 3f). The ratios of BG/NAG and BG/ALP were significantly elevated under long-term *T. repens* but remained unchanged under other treatments (Fig. 3d and h).

Compared to the control, long-term *V. villosa* significantly increased bacterial observed species, whereas other treatments generally decreased bacterial observed species, Shannon, and Simpson indices—with long-term *T. repens* showing the most pronounced reduction (Table 3). Cover crops significantly reduced fungal observed species but increased fungal Shannon and Simpson indices.

Among treatments, long-term *T. repens* demonstrated the strongest positive effect on fungal diversity indices (Table 3).

3.6 Influence of soil microbial properties on soil organic carbon fractions

The RDA results showed that soil microbial properties explained 97.1% of the variation in SOC physical fractions (Fig. 4a). The MBC, bacterial Chao1, BG/NAG, PHO and FNC were identified as key explanatory factors (Fig. 4b). The MBC, BG/NAG and FNC exerted positive effects on both POC and MAOC, while The PHO negatively influenced both POC and MAOC. The bacterial Chao1 exerted a positive effect on MAOC and a negative effect on POC (Fig. 4a).

The PLS-PM results indicated that the increased DOC caused by short-term cover crops positively affected MBC, and indirectly affected the MAOC and SOC (Fig. 5a). The fungal community diversity had positive effects on enzyme activity, and the enzyme activity was negatively correlated with POC (Fig 5a). Under long-term cover crops, increased soil DOC positively correlated with the fungal community diversity and MBC, which positively affected the FNC but negatively affected with enzyme activity (Fig 5b). Under this treatment, strong positive relationships were observed between FNC and MAOC and between MAOC and SOC (Fig 5b). Additionally, enzyme activity and POC were positively correlated under long-term treatments.

4. Discussion

4.1 Short-term cover crops fail to enhance SOC due to reduced POC

The sequestration of SOC by cover crops in orchard agroecosystems has attracted considerable attention (Ball et al., 2020a; Zhang et al., 2022; Vendig et al., 2023; Acharya et al., 2024). Contrary to the prevailing consensus on its positive effects on SOC and associated fractions (Hu et al., 2024), our study revealed a distinct short-term dynamic: cover crops significantly reduced POC content,

while exerting either neutral or marginally stimulatory effects on MAOC. This divergence in carbon fraction responses culminated in an overall decline or non-significant alteration of total SOC stocks under short-term treatment.

POC, primarily derived from the physical fragmentation of plant residues or other labile organic matter, making it sensitive to cover crops (Blanco-Canqui et al., 2017). Cover crops enhance the accumulation of particulate organic carbon (POC) by increasing inputs of plant residues and root exudates, which are physically occluded within soil aggregates (Wu et al., 2014; Yang et al., 2022; Hu et al., 2023b). However, labile carbon inputs from cover crops may also stimulate microbial activity, thereby accelerating the decomposition of native SOC via priming effects (PE) (Dijkstra et al., 2021; Xu et al., 2024). Since POC lacks mineral protection and represents the most labile SOC fraction (Cotrufo and Lavelle, 2022), it is highly susceptible to decomposition under external perturbations (Duval et al., 2010; Pisani et al., 2015). The observed increases in polyphenol oxidase (PPO) activity in our study (Fig. 3), and its significant negative relationship with POC (Fig S2 and Fig. 5), indicate intensified decomposition of both newly added or native SOC. Consequently, despite the theoretical potential for POC accumulation, cover crops induced significant POC reduction-driven by enhanced priming effects (PE) and limited plant-derived inputs under short-term application-culminating in net carbon loss. In addition, the observed POC decline in our study may also be attributed to the sandy soil's inherent limitations-compared to clay and silt-dominated soils, sands exhibit less capacity to form stable aggregates, providing less effective SOC protection via microbial isolation (Six et al., 2004; Gyawali and Stewart, 2019).

MAOC, mainly originating from microbial residues that bind to mineral surfaces forming stable structures, has a prolonged turnover time relative to POC (Cotrufo and Lavelle, 2022;

Lavallee et al., 2020). While previous studies have demonstrated that cover crops enhanced MAOC by increasing the microbial-derived residues (Duval et al., 2016; Zhang et al., 2022; Hu et al., 2023a), our findings revealed a contrasting short-term dynamic: cover crops exerted no significant effect on MAOC under *T. repens* and even significantly reduced MAOC under *V. villosa*. Our result was partly accordance with Castro et al. (2015) and Martinez et al. (2020) found that unchanged MAOM under cover crops application. The delayed accrual of MAOC stems from the time-dependent progression from microbial growth to necromass stabilization, as microbial residues require extended turnover periods to integrate into mineral-bound carbon pools (Cotrufo and Lavallee, 2022; Liang et al., 2017). Our study revealed that short-term cover crops increased soil microbial biomass while maintaining necromass carbon levels under *T. repens*, indicating preferential allocation of plant-derived carbon toward microbial growth and extracellular enzyme production over necromass accumulation during initial stages. What's more, the significant decrease in MAOC under short-term *V. villosa* is positively correlated with the reduction of microbial residue carbon (Fig.S2). As previous study documented that residues with a higher C:N ratio and lignin leading to stronger PE (Mazzilli et al., 2014), and forced microorganisms to mineralize more N and nutrients from mineral-associated organic matter. *V. villosa* with a higher C:N ratio and lignin compared to *T. repens* may trigger stronger PE effect suggested by the highest PPO (Fig. 3f), increased the new or native MAOC decomposition, thereby resulting MAOC reduction.

4.2 Long-term cover crops enhance SOC simultaneously increasing POC and MAOC fractions

In contrast to short-term effect, long-term cover crops significantly enhanced SOC content, a finding consistent with previous studies (Jian et al., 2020; Zhang et al., 2022; Hu et al. 2023a). However, species-specific divergence occurred: *T. repens* stimulated both soil fractions accrual,

particularly POC, yielding higher SOC than *V. villosa* treatment; conversely, *V. villosa*-driven SOC increases were primarily attributed to MAOC enrichment, with the POC had no significant increase.

The soil microbial ex vivo pathway plays a critical role in plant-derived carbon formation and accumulation (Liang et al., 2017; Hu et al., 2024). Our study reveals that cover crops with contrasting chemical qualities differentially regulate this pathway through soil enzymatic strategies, ultimately governing POC accumulation dynamics. Compared to *V. villosa* with a higher C/N ratio and lignin content, *T. repens* exhibited superior chemical quality, with a lower C/N ratio and greater non-structural carbohydrate content (soluble sugars and starch). These labile, carbohydrate-rich components likely provide more essential substrates (e.g., DOC) for microbial metabolism, supporting the synthesis of C degrading enzymes (Ball et al., 2020; Kim et al., 2020; Ding et al., 2024), as evidenced by the significantly higher CBH activity relative to *V. villosa* (Fig.3). The stimulation of enzymes for C and N cycling increase microbial energy enhances the decomposition of plant macromolecules into small fragments, drives more efficient conversion of plant residues into stable POC. Additionally, *V. villosa* intercropping significantly increases NAG activity, resulting in a lower BG/NAG ratio. This reduced ratio may trigger a microbial N-mining mechanism, leading to a positive priming effect and accelerating the decomposition of POC (Liang et al., 2023). Furthermore, peroxidase activity—critical for decomposing recalcitrant organic matter—is markedly higher under *V. villosa* than *T. repens*, likely further exacerbating POC mineralization. These enzymatic synergies ultimately widen the disparity in net POC accumulation between the two species. Consequently, despite the potential for increased POC accumulation suggested by more plant litter input (Table 1), significant POC accumulation was not detected under *V. villosa*.

Differences in the quality of legume cover crops also affected MAOC formation, as reported

in other studies (Hu et al., 2023a), through the in vivo pathway. According to the Microbial Efficiency-Matrix Stabilization framework (Cotrufo et al., 2013), the higher the quality induce the higher the microbial substrate use efficiency, and the higher the build-up of microbial biomass, whose residues eventually interact with the soil mineral phase to form MAOM. However, Contrary to the aforementioned theories, Our study results revealed that soil MAOC content under low-quality *V. villosa* was significantly 20% higher than under high-quality *T. repens* (Fig.1). It is widely accepted that microbial necromass carbon, particularly fungal necromass C, was the key driver of MAOC accumulation under cover crops (Asghar and Kataoka, 2022). Consistent with previous studies, the application of two cover crop types showed no significant effects on bacterial necromass carbon content, but significantly promoted fungal necromass carbon accumulation, resulting in a marked increase in total microbial necromass carbon. This may represent one of the underlying mechanisms through which cover crop enhances soil mineral-associated organic carbon (MAOC). However, under low-quality *V. villosa*, fungal and microbial necromass carbon content was lower (though not statistically significant) compared to *T. repens*, a pattern inconsistent with its observed higher MAOC formation. This discrepancy implies that MAOC accumulation may not exclusively originate from microbial sources, but could also be partially derived from plant residues. This guess are supported by prior studies by others that plant-derived carbon compounds predominately account for MAOC pool relative to microbial-derived carbon compounds (Angst et al., 2021). The present study revealed that long-term intercropping with *V. villosa* (containing higher lignin and acid-insoluble substances) enhanced soil peroxidase and phenol oxidase activities by 20% compared to the control and 30% compared to *T. repens* (Fig. 3f and g). These oxidative enzymes decomposed recalcitrant plant residues (e.g., lignin, phenols, and cutin) (Men et al., 2024), promoting their

binding with minerals to form more stable mineral-associated organic carbon (MAOC) (Sanderman et al., 2014). The elevated oxidative enzyme activity under *V. villosa* likely contributed to MAOC accumulation through this mineral-organic interaction mechanism, which is according with the positive relationship with oxidative enzymes with MAOC (Fig. 5b).

4.3 The complex relationship between microbial properties and soil organic carbon sequestration

Clarifying how microbial properties govern SOC sequestration remains a key challenge in soil ecology (Bhattacharyya et al., 2022), which was a primary objective of this study. The RDA results revealed that soil microbial properties significantly explained the variation in SOC physical fractions. We found strong positive relationships between MAOC and MBC and fungal necromass C, indicating the higher metabolic efficiency of fungi in the microbial in vivo carbon processing pathway enhances fungal necromass production. Subsequent Synergistic interactions between this fungal necromass carbon (C) and mineral surfaces further promote mineral-associated organic carbon (MAOC) formation. In addition, key C decomposers such as polyphenol oxidase, which is known for its ability to catalyse phenolic and lignin compounds significantly and negatively contributed to changes in POC, and BG:NAG ratio showed significantly positive relationships with POC. These associations can be explained by: a) low quality substrates promoted polyphenol oxidase activities induced higher priming effect on non-stable POC, therefore decreased POC. b) increased soil N availability indicated microbial N starvation, which may even trigger a microbial N-mining mechanism, leading to a positive priming effect and accelerating the decomposition of POC.

4.4 Limitations and uncertainties

This study has several limitations and uncertainties. First, while microbial residue carbon (MBN) was quantified in bulk soil, its distribution across mineral-associated organic matter

(MAOM) or particulate organic matter (POM) fractions remains unexamined. Clarifying the plant- versus microbial-derived origins of POC and MAOC is essential to refine mechanistic understanding of their formation and transformation pathways under cover crops. Second, the impact of cover crops on soil organic carbon represents a balance between priming effects versus burial effects (Dijkstra et al., 2021). Employing stable isotope tracing to quantify the conversion of cover crop-derived carbon into microbial necromass carbon (MNC) and its subsequent storage as mineral-associated organic carbon (MAOC) is critical to understand the complex linkages between microbial traits and SOC fractions under cover crops. Additionally, the geographically restricted findings (Nyingchi, Tibet; two cover crops: *V. villosa* and *T. repens*) necessitate broader validation, as SOC sequestration patterns likely vary across agroecological zones and with different cover crop species.

5. Conclusion

In this study, we combined SOC fractions and microbial properties to investigate changes in SOC sequestration under different cover crops. We conclude that short-term legume cover crops exerted no significant effect on MAOC but markedly reduced POC, resulting in negligible or even decreased total SOC content. Long-term cover crops significantly enhanced both total SOC and its fractions. *T. repens* preferentially promoted the accumulation of POC, reducing the MAOC/POC ratio and compromising SOC stability. In contrast, although long-term cover by *V. villosa* was less effective in increasing total SOC than *T. repens*, it selectively enriched MAOC, thereby enhancing SOC stability. Collectively, our analyses provide insights into the divergent efficiencies of cover crops in enhancing SOC sequestration across species and over time, providing a scientific foundation for formulating effective legume cover cropping strategies.

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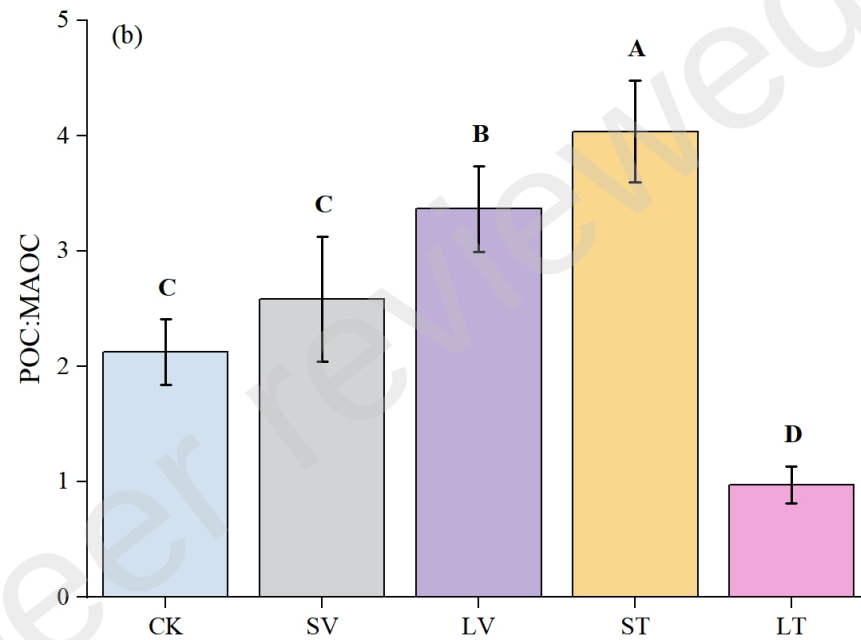
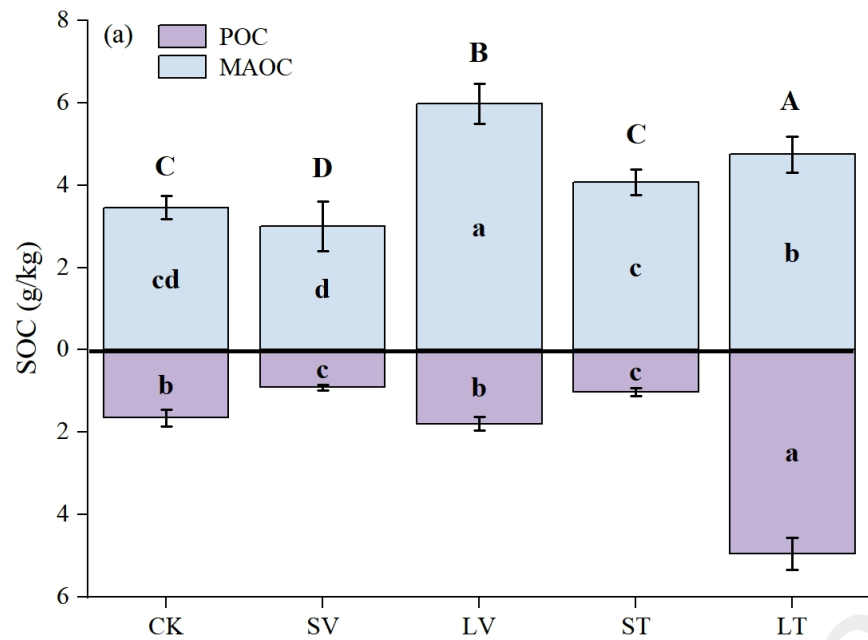
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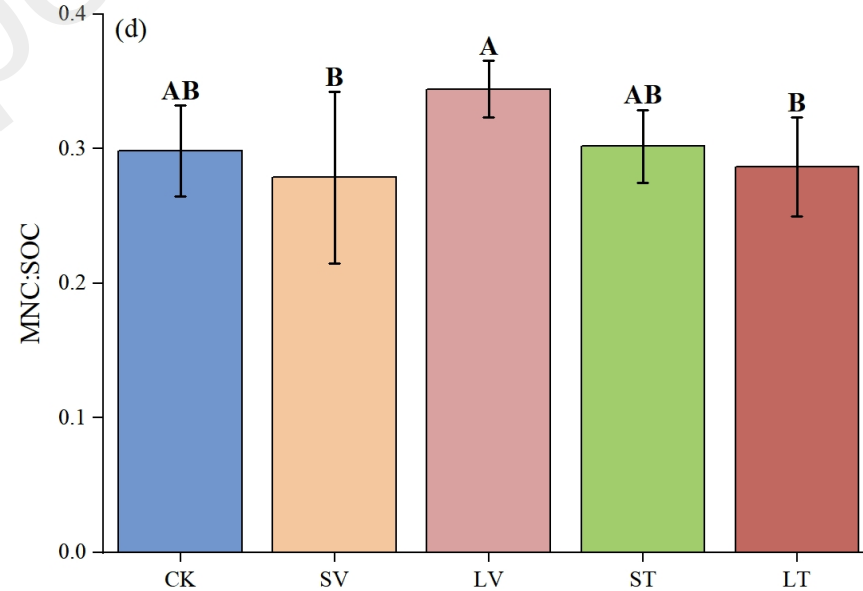
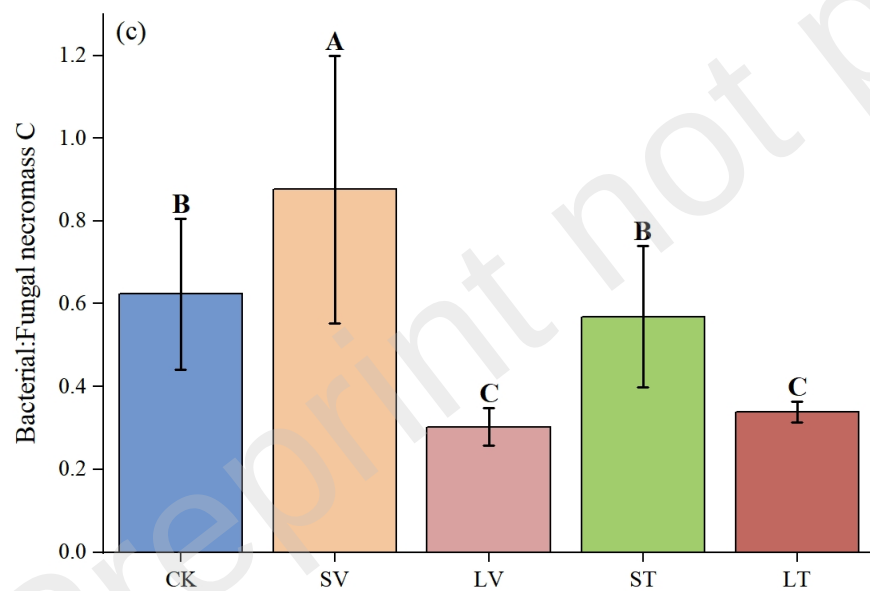
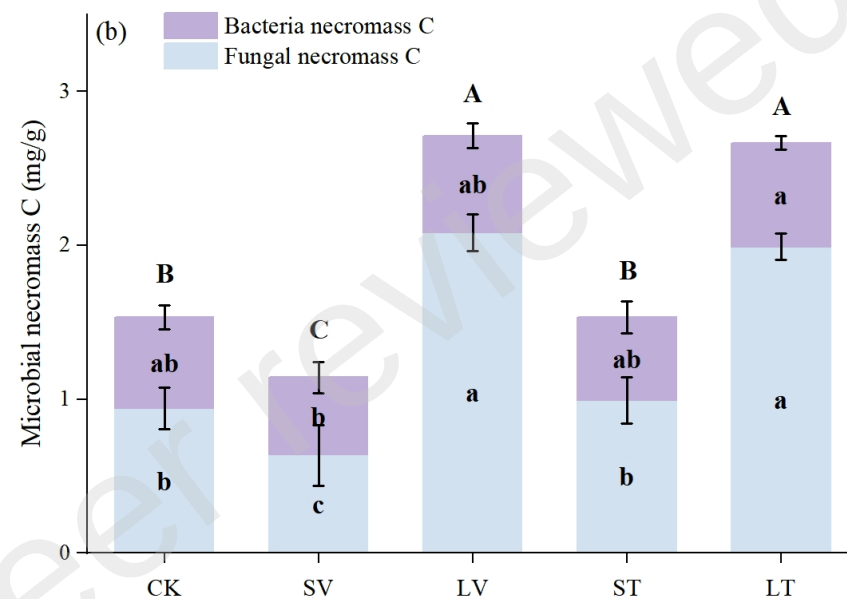
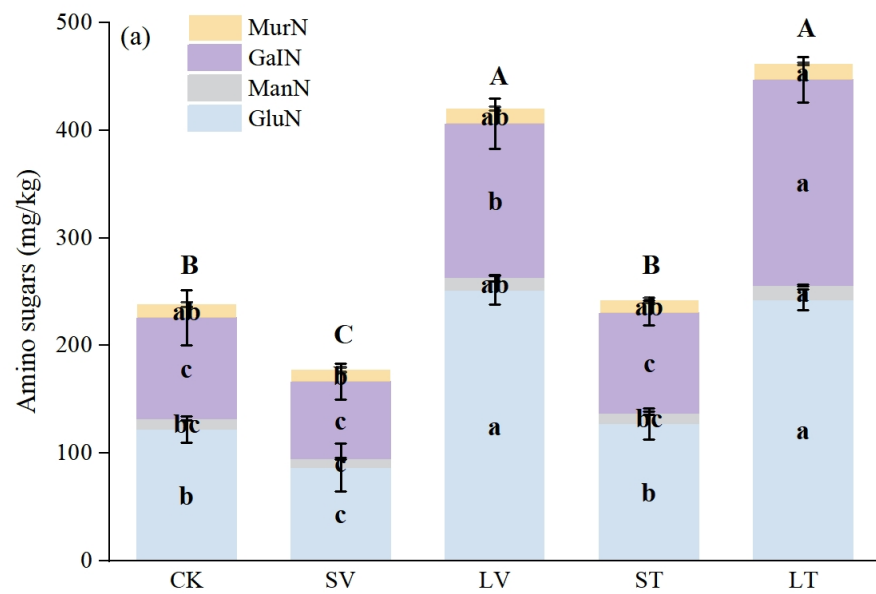
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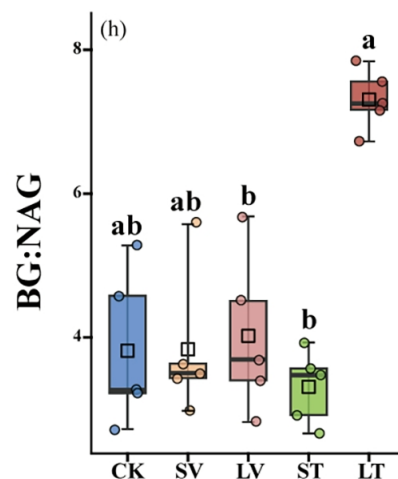
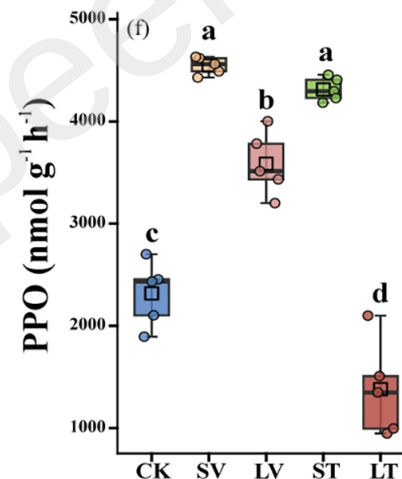
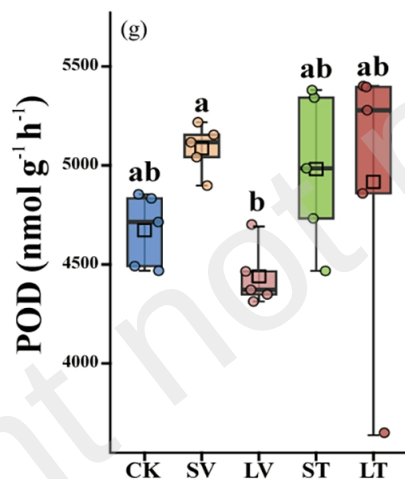
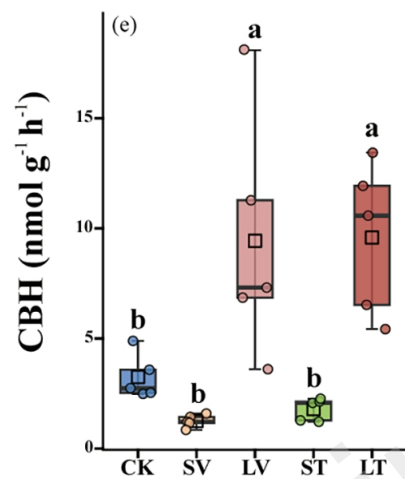
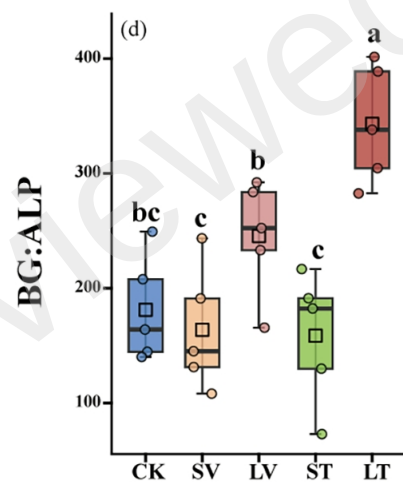
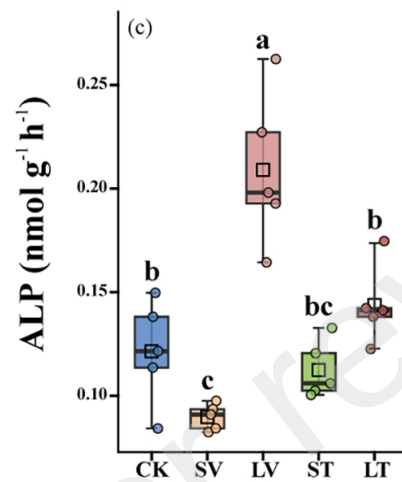
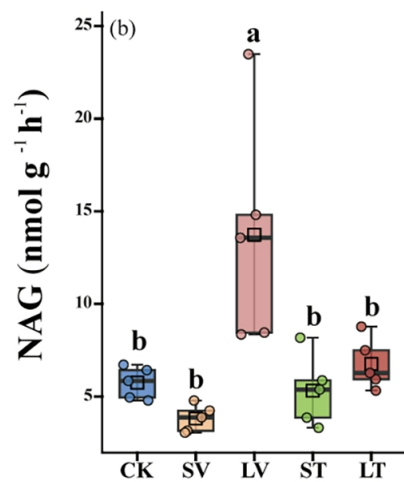
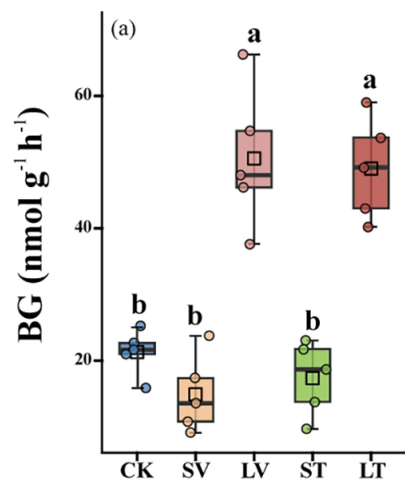
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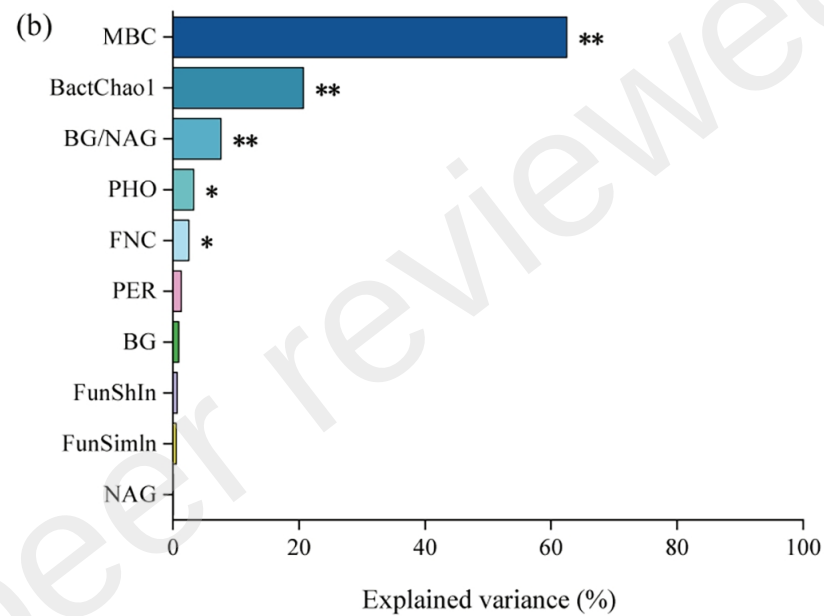
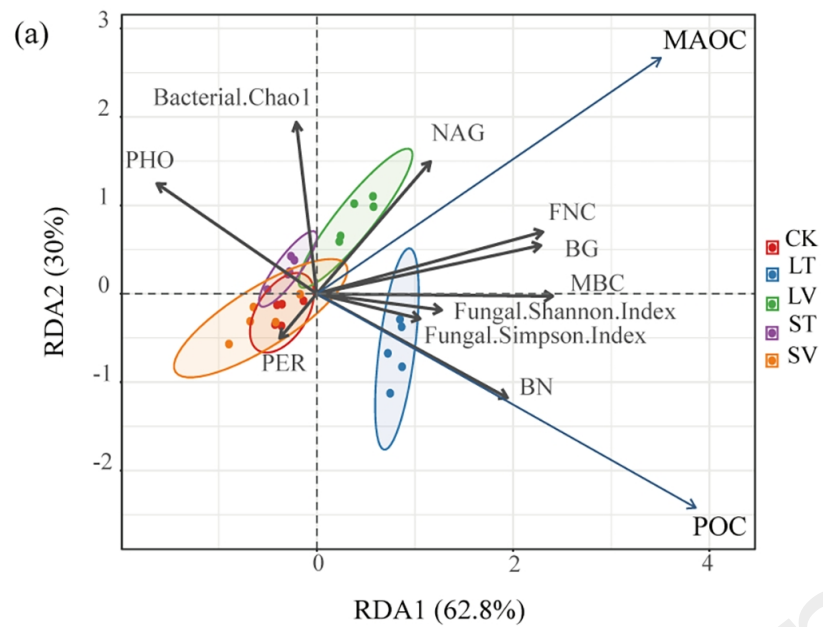
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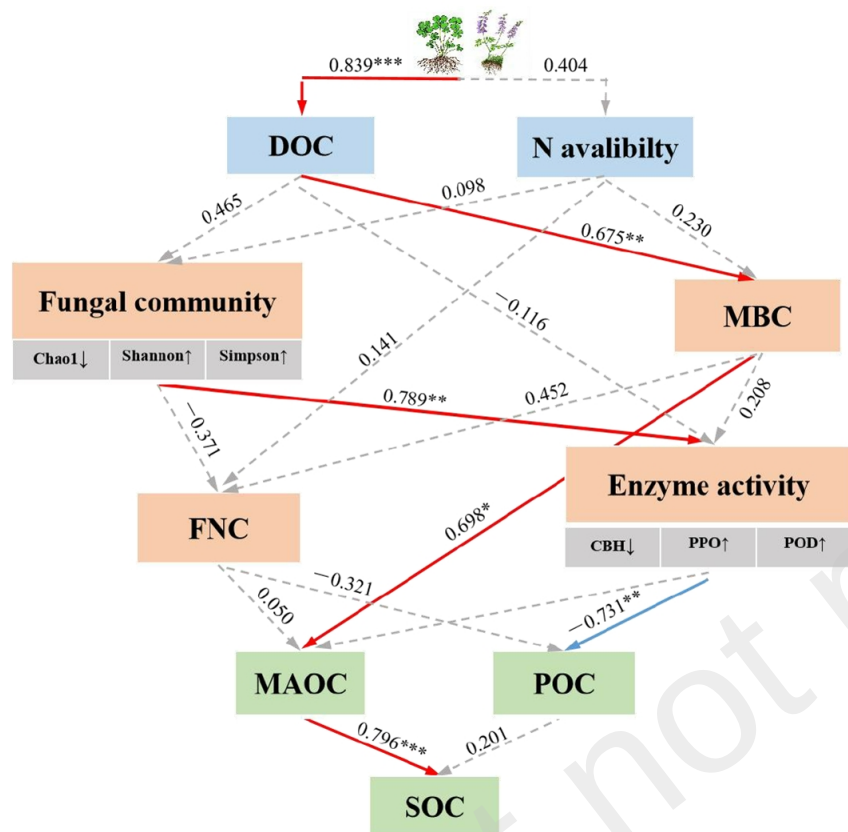








(a) Short-term cover crops



(b) Long-term cover crops

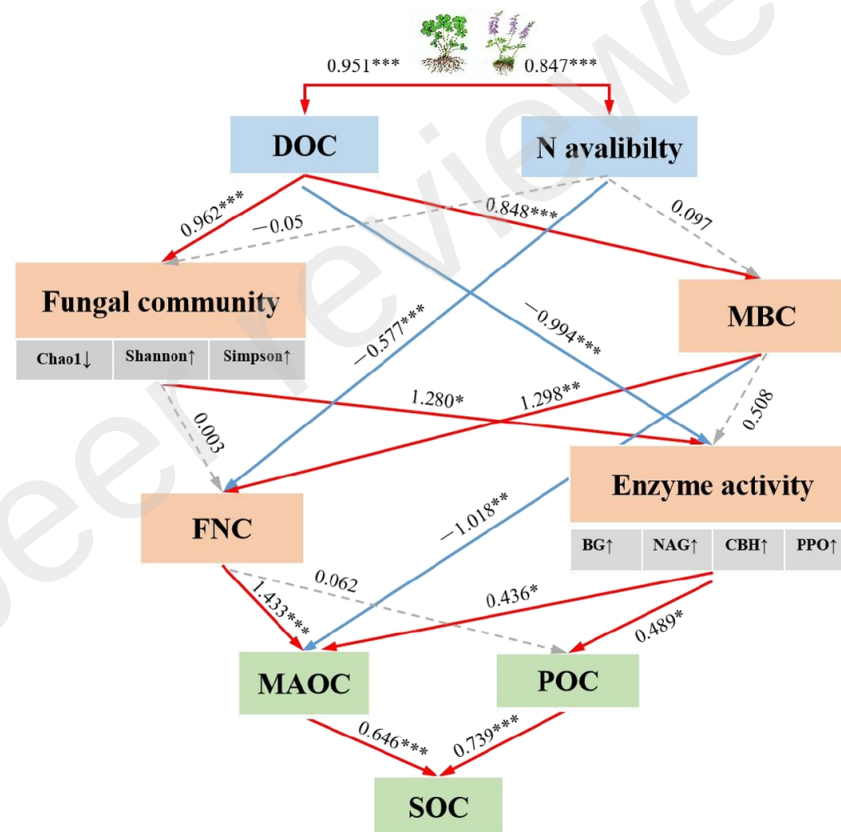


Figure 1 Responses of soil organic carbon (SOC) fractions to different cover crops. Lowercase letters denote significant differences between treatments ($p < 0.05$). Bars represent the means $\pm$ standard deviation ($n = 5$). POC, particulate organic carbon; MAOC, mineral-associated organic carbon.

Figure 2 Responses of soil microbial necromass to different cover crops. Lowercase letters denote significant differences between treatments ($p < 0.05$). Bars represent the means $\pm$ standard deviation ($n = 5$). MurA, muramic acid; ManN, amino mannose; GluN, glucosamine; GalN, galactosamine; C, carbon; MNC, microbial necromass carbon.

Figure 3 Responses of soil enzyme activities to different cover crops. Lowercase letters denote significant differences between treatments ($p < 0.05$). Bars represent the means $\pm$ standard deviation ($n = 5$). BG, β -glucosidase; CBH, cellulose hydrolase; NAG, β -N-acetylglucosaminidase; ALP, alkaline phosphatase; POD, peroxidase; PPO, polyphenol oxidase.

Figure 4 Redundancy analysis (RDA) of soil organic carbon (SOC) fractions constrained by microbial properties (a) and the variance in microbial properties (b). POC, particulate organic carbon; MAOC, mineral-associated organic carbon; MBC, microbial biomass carbon; PPO, polyphenol oxidase; BG, β -glucosidase; NAG, β -N-acetylglucosaminidase; FNC, fungal necromass carbon.

Figure 5 A directed graph of the partial least squares path model (PLS-PM) shows the effects of cover crops, soil nutrients, and microbial properties on soil organic carbon (SOC). The numbers adjoining the lines indicate standardized path coefficients. Red lines represent positive relationships; blue lines represent negative relationships. The solid lines represent significant relationships; the dashed lines represent nonsignificant relationships. Significance is represented by assessing the

degree of model fit: *** when $p < 0.001$, ** when $p < 0.01$, and * when $p < 0.05$. N, available nitrogen; DOC, soil-dissolved organic carbon; POC, particulate organic carbon; MAOC, mineral-associated organic carbon; MBC, microbial biomass carbon; MNC, microbial necromass carbon; GoF, goodness-of-fit.

Table 1 Legume cover biomass and chemical properties

Treatments	Above ground biomass (g/m ²)	TC (g/kg)	TN (g/kg)	TP (g/kg)	SS %	Starch %	Cellulose %	Lignin %	C/N	C/P
SV	1839.45±80.61b	440.61±1.02a	34.29±2.16ab	2.38±0.16b	4.43±0.21b	10.82±0.80a	23.04±3.98bc	5.41±0.28b	12.90±0.79a	18.62±1.16a
LV	2119.55±76.61a	411.56±1.60c	30.65±1.61bc	3.98±0.21a	1.85±0.18c	8.42±0.62b	34.35±5.37a	6.37±0.80a	13.47±0.72a	10.38±0.56b
ST	1930.90±60.78ab	420.01±4.00b	30.34±1.15c	2.39±0.10b	4.55±0.26b	8.48±0.46b	26.74±1.11b	4.13±0.15c	13.87±0.56a	17.59±0.82a
LT	1327.35±215.61c	424.02±2.17b	36.34±1.65a	2.52±0.24b	9.03±0.33a	9.85±0.54a	19.45±0.99c	2.87±0.32d	11.70±0.49b	17.00±1.66a

TC, total carbon; TN, total nitrogen; TP, total phosphorus; SS, soluble sugar; AUF, acid-unhydrolyzable fraction. Values are the mean ± standard deviation (n = 5). Lowercase letters denote significant differences between treatments (p < 0.05).

Table 2 Responses of soil properties to legume cover crops

Treatments	DOC (mg/kg)	AN (mg/kg)	NH ⁴⁺ (mg/kg)	NO ³⁻ (mg/kg)	AP (mg/kg)	MBC (mg/kg)	MBN (mg/kg)
CK	34.17±2.86d	5.93±1.18b	0.39±0.15c	5.78±5.06b	3.02±4.62c	36.68±1.67d	3.29±0.73c
SV	37.40±4.92d	7.12±0.74b	0.48±0.02bc	7.24±0.67b	3.97±0.76c	35.42±4.18d	4.01±1.27c
LV	57.27±6.65b	7.17±1.36b	0.62±0.18b	6.55±1.17b	47.02±12.86b	91.34±4.96b	12.89±1.07a
ST	48.10±2.53c	9.82±6.82a	0.51±0.09bc	9.30±6.10a	11.81±2.87c	55.15±3.14c	6.16±0.21b
LT	79.25±9.04a	17.61±3.57a	1.25±0.28a	16.36±3.27a	124.08±21.76a	122.02±4.29a	14.02±0.99a

DOC, dissolved organic carbon; AN, available nitrogen; NH⁴⁺, ammonium N; NO³⁻, nitrate N; AP, available phosphorus; MBC, microbial biomass carbon; MBN, microbial biomass nitrogen.

Values are the mean ± standard deviation (n = 5). Lowercase letters denote significant differences between treatments (p < 0.05).

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Table 3 Responses of soil microbial community diversity to legume cover crops

Treatments	Bacterial			Fungal		
	Observed_species	Shannon	Simpson	Observed_species	Shannon	Simpson
CK	5919.5±990.89b	5.4478±0.19a	0.9745±0.007a	83.3±16.75a	1.225±0.09c	0.4282±0.02c
SV	5401.9±310.72b	4.6757±0.16ab	0.932±0.007bc	59.0±3.79b	1.6467±0.03ab	0.5638±0.008ab
LV	6995.9±517.80a	4.5846±0.07ab	0.9417±0.004b	61.5±5.36b	1.6384±0.007ab	0.5369±0.005b
ST	5287.1±282.52b	4.6953±0.24b	0.9312±0.017bc	59.4±6.08b	1.5262±0.28b	0.5294±0.07b
LT	4416.6±237.21c	4.4358±0.18c	0.9185±0.012c	56.8±13.4b	1.8334±0.24a	0.5946±0.04a

Values are the mean ± standard deviation (n = 5). Lowercase letters denote significant differences between treatments (p < 0.05).